

Chapter 4 — Output Stages and Power Amplifiers

Electronics Device and Circuit (ENEX 151 / EX 151) · Tribhuvan University, IOE · BE (BEI, BCT) · Year I / Part II

Exam-oriented notes with embedded Past-Year Questions (2081 Ashwin R · 2082 Baishakh B · 2082 Bhadra R · 2083 Baishakh B)

References: Boylestad (efficiency formulas — every PYQ numerical follows his method), Sedra-Smith (output-stage classification, class-AB biasing, V_{BE} multiplier), Millman, Floyd

Figures: 11 diagrams rendered locally — load lines, waveforms and power curves computed from the actual equations with the PYQ parameter values (fetch report: 0/11 from web)

How to Use These Notes / Chapter Overview

Power amplifiers deliver **watts to real loads** — speakers, antennas, motors — and the figure of merit flips from gain to **conversion efficiency** $\eta = P_o(ac)/P_i(dc)$, always traded against distortion. The whole chapter is a ladder of **classes** defined by conduction angle: A (linear, wasteful, 25/50%) → B (efficient 78.5%, but crossover-distorted) → AB (the practical compromise) → C/tuned for RF. Every PYQ is either an efficiency *derivation* or an efficiency *numerical*, all straight from Boylestad's chapter.

PYQ Frequency Analysis — Chapter 4 is worth 12-13 marks every single year

Topic	2081 Ashwin	2082 Baishakh	2082 Bhadra	2083 Baishakh	Verdict
Class B: circuit / efficiency derivation / numerical	✓ push-pull η derivation [5]	✓ numerical, maxima [5]	✓ transformer circuit + η condition [6]	✓ numerical [6]	★ 4/4 GUARANTEED
Transformer-coupled Class A (circuit, curves, η)	✓ numerical η [4]	-	-	✓ general + max η [7]	★ 2/4
Series-fed Class A η derivation	-	✓ [4]	-	-	1/4
Class AB / crossover distortion	-	-	✓ [6]	-	1/4
Tuned amplifier (circuit, response, bandwidth)	✓ 3 dB BW [4]	✓ BW derivation [4]	-	-	★ 2/4
Classification / power BJTs / heat	implied	-	-	-	readiness only

Strategic conclusions

- ★ **Class B is the only 4/4 guarantee in the whole paper besides the FET pair** — and it alternates between derivation years and numerical years. Prepare BOTH templates: the 8-line η derivation and the $P_i/P_o/\eta$ + maxima numerical.
- ★ Transformer-coupled Class A and the tuned amplifier each hit 2/4 — a derivation and a numerical each. Together with Class B they cover ≈ 11 of the 13 marks in every past paper.
- The chapter's arithmetic is trivial; the marks live in the *setup lines*: which power formula belongs to which class, and the three magic results 25% / 50% / 78.5%. The classic slip — using a Class A formula in a Class B numerical — costs the whole question.

4.1 Classification of Output Stages

Classes are defined by the **conduction angle** — the fraction of each signal cycle for which the output device carries current. Sliding the Q-point down the load line from centre (A) to cutoff (B) and below (C) trades linearity for efficiency:

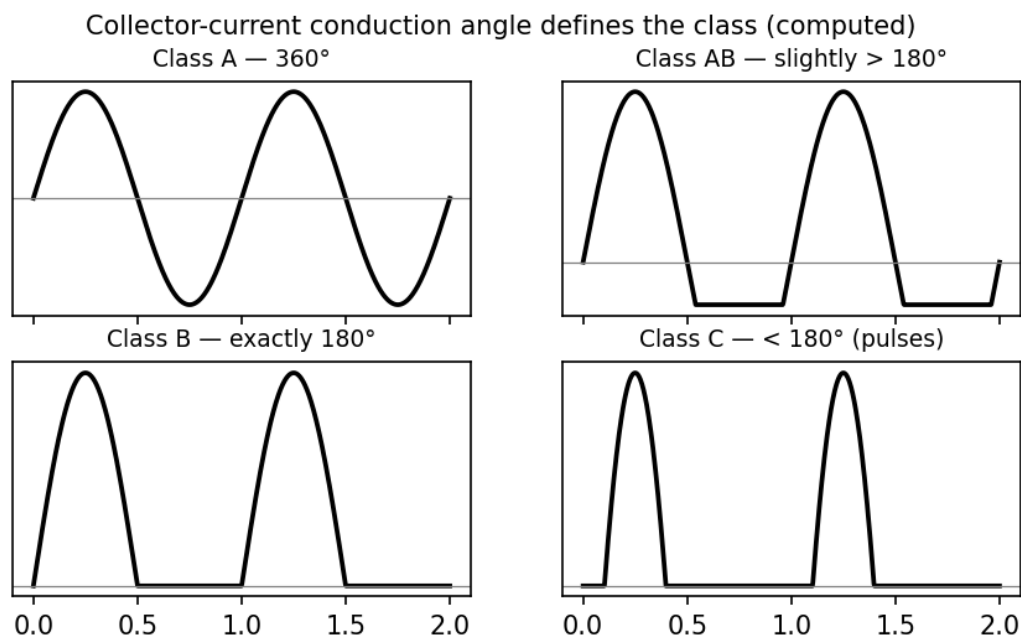


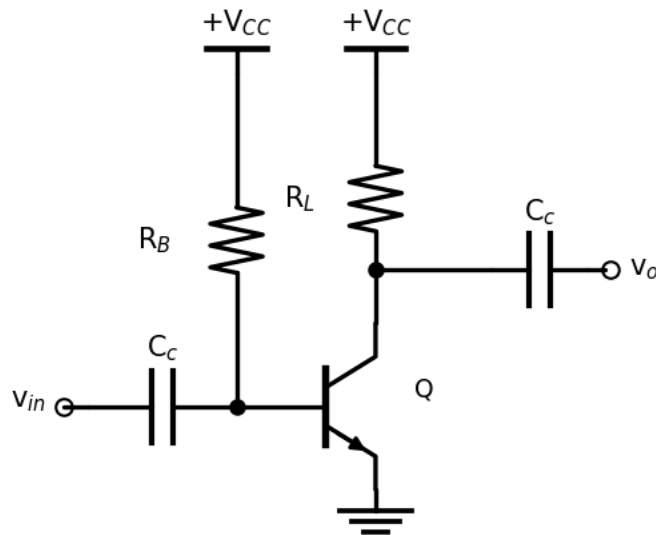
Fig. 1 — Collector current over two cycles for each class, computed by clipping a sinusoid at the respective bias level. [computed]

Class	Conduction angle	Q-point	Max η	Distortion	Typical use
A	360°	centre of load line	25% (series) / 50% (transformer)	lowest	small-signal, hi-fi driver stages
B	180°	at cutoff	78.5%	crossover notch	push-pull output pairs
AB	just over 180°	slightly above cutoff	between B and A	low (notch removed)	practical audio output
C	< 180°	below cutoff	> 90%	severe (pulses)	RF with tuned tank load

MEMORY HOOK: Exam one-liner that frames any Chapter-4 answer: **as the conduction angle falls, efficiency rises and linearity falls** — the entire chapter in one sentence.

4.2 Class A Output Stages

A. Series-fed Class A — PYQ 2082 Baishakh Q9 [4]: derive the general efficiency



load R_L sits directly in the collector — it carries I_{CQ} at DC

Fig. 2 — Series-fed Class A: an ordinary CE stage with the load directly in the collector — R_L carries the full quiescent current at DC. [rendered locally]

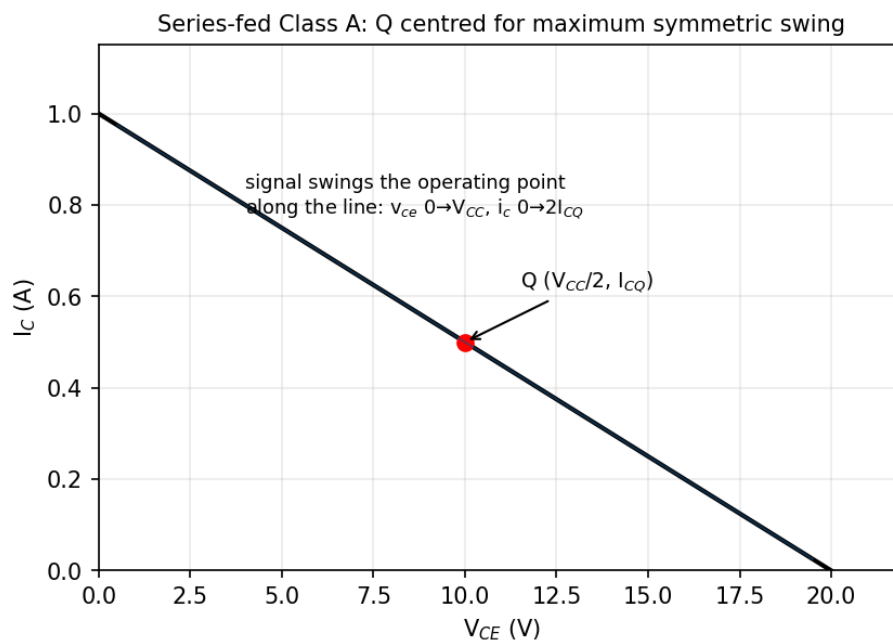


Fig. 3 — Q is placed mid-load-line ($I_{CQ} = V_{CC}/2R_L$) so the signal can swing symmetrically: v_{CE} from 0 to V_{CC} , i_C from 0 to $2I_{CQ}$. [computed]

Derivation (write all four steps)

- **1. DC input power:** the supply delivers the constant bias current whatever the signal does — the sine's average is zero — so $P_i(\text{dc}) = V_{CC} \cdot I_{CQ}$.
- **2. AC output power** for output peaks V_p (voltage) and I_p (current): $P_o(\text{ac}) = V_p I_p / 2 = V_p^2 / (2R_L)$ (rms product).
- **3. General efficiency:** $\eta = P_o / P_i = (V_p I_p / 2) / (V_{CC} I_{CQ}) \times 100\%$.
- **4. Maximum:** the largest undistorted swing is $V_p = V_{CC}/2$ and $I_p = I_{CQ} = V_{CC}/2R_L \Rightarrow P_o(\text{max}) = V_{CC}^2 / 8R_L$ while $P_i = V_{CC}^2 / 2R_L$:

$$\eta_{\text{max}} = (V_{CC}^2 / 8R_L) / (V_{CC}^2 / 2R_L) = 1/4 = 25\% \blacksquare$$

- **Weakness to state for the last mark:** dissipation is *worst at zero signal* ($P = V_{CC}I_{CQ}$ burnt with nothing delivered), and half the supply power is wasted as DC heat in R_L itself — which motivates the transformer.

B. Transformer-coupled Class A ★ REPEATED PYQ TOPIC — PYQs 2083 Q8 [7] (circuit + curves + general && max η) and 2081 Q9 [4] (numerical)

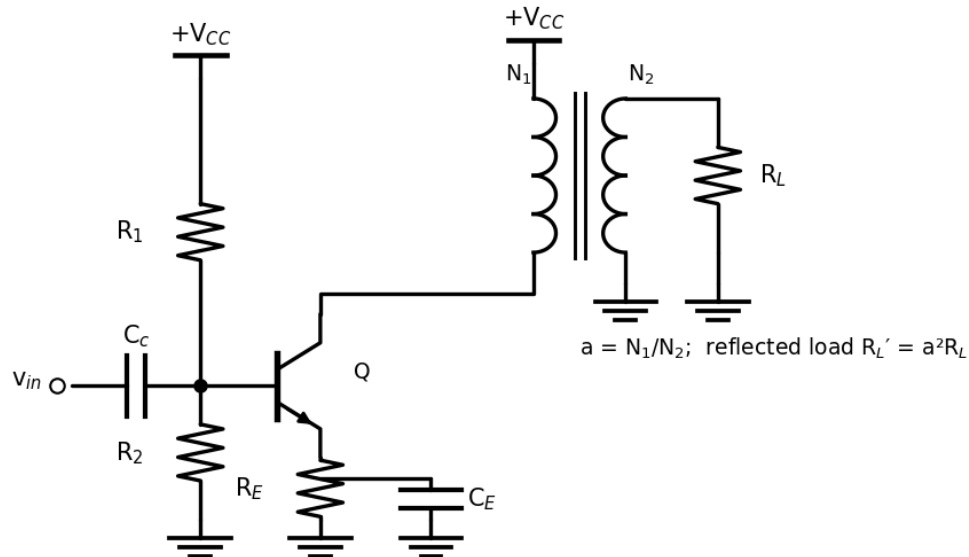


Fig. 4 — Transformer-coupled Class A: the collector load is the primary of a step-down transformer (turns ratio $a = N_1/N_2$); the actual load R_L on the secondary appears at the collector as $R_L' = a^2 R_L$. The winding is $\approx 0 \Omega$ at DC — no DC power wasted in the load path. [rendered locally]

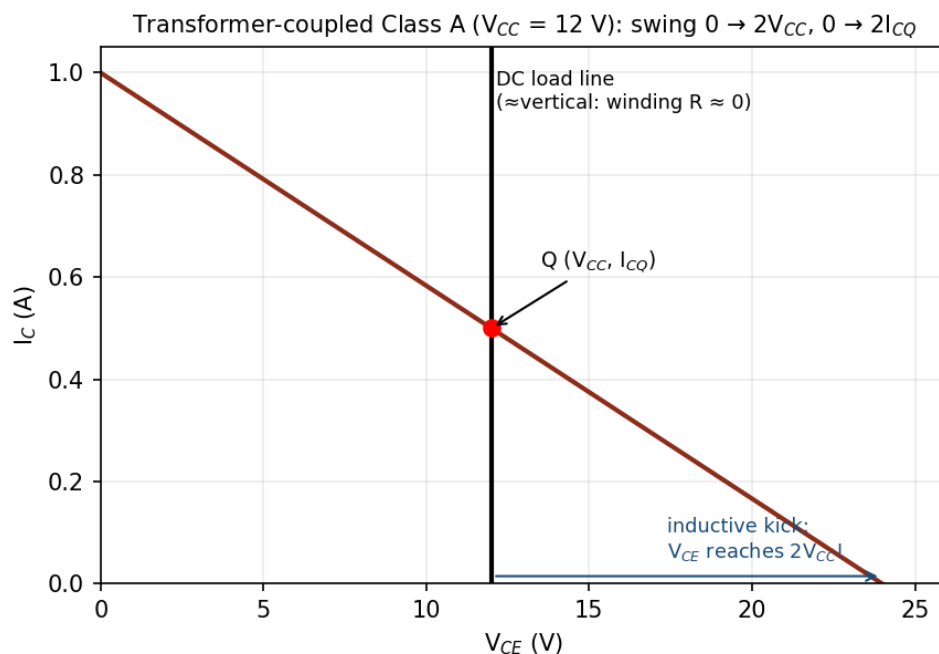


Fig. 5 — The characteristic-curve sketch the 7-marker demands: DC load line essentially VERTICAL at V_{CC} (zero winding resistance); AC load line of slope $-1/R_L'$ through Q . The inductive kick carries V_{CE} above the rail — the swing reaches $2V_{CC}$. [computed ($V_{CC} = 12 \text{ V}$, the 2081 Q9 supply)]

General-efficiency derivation [the 7-marker's core]

1. $P_i(\text{dc}) = V_{CC} \cdot I_{CQ}$ — constant, exactly as before.
2. With swings between $(V_{CE\text{max}}, I_{C\text{min}})$ and $(V_{CE\text{min}}, I_{C\text{max}})$, write the AC power in peak-to-peak form: $P_o(\text{ac}) = (V_{CE\text{max}} - V_{CE\text{min}})(I_{C\text{max}} - I_{C\text{min}})/8$ (each peak amplitude is half the

peak-to-peak; the $\frac{1}{2}$ of rms² brings the 8).

- **3.** Divide, using the symmetric-swing facts $V_{CEmax} + V_{CEmin} = 2V_{CC}$ and $(I_{Cmax} - I_{Cmin}) = (V_{CEmax} - V_{CEmin})/R_L$, $I_{CQ} = (I_{Cmax} + I_{Cmin})/2$:

$$\eta = 50 \times [(V_{CEmax} - V_{CEmin}) / (V_{CEmax} + V_{CEmin})]^2 \% \quad (\text{Boylestad's general form — equivalently } 50 \cdot (V_p/V_{CC})^2 \%)$$

- **4. Maximum:** full swing $V_{CEmax} = 2V_{CC}$, $V_{CEmin} = 0 \Rightarrow \eta_{max} = 50 \times (2V_{CC}/2V_{CC})^2 = \mathbf{50\%}$ ■ — double the series-fed value *because no DC power is burnt in a collector resistor* (the primary is $\approx 0 \Omega$ at DC). Say that sentence; it is a mark.

★ **Worked PYQ — 2081 Ashwin Q9 [4]:** $V_{CC} = 12 \text{ V}$; efficiency for output peaks $V_p = 12 \text{ V}$ and 6 V

- **$V_p = 12 \text{ V}$:** $V_{CEmax} = 12 + 12 = 24 \text{ V}$, $V_{CEmin} = 12 - 12 = 0 \text{ V} \Rightarrow \eta = 50 \times (24/24)^2 = \mathbf{50\%}$ — the theoretical maximum (full swing).
- **$V_p = 6 \text{ V}$:** $V_{CEmax} = 18 \text{ V}$, $V_{CEmin} = 6 \text{ V} \Rightarrow \eta = 50 \times (12/24)^2 = 50 \times 0.25 = \mathbf{12.5\%}$.
- **Observation (the last mark):** at half drive the efficiency has fallen to a quarter — Class A efficiency collapses *quadratically* with signal level: $\eta = 50(V_p/V_{CC})^2\%$.
- ⚠ Mistakes: drawing the DC load line with slope $-1/R_L$ (it is vertical here) · capping the swing at V_{CC} (transformer action reaches $2V_{CC}$) · quoting 25% as this circuit's maximum.

4.3 Class B Output Stage ★ REPEATED PYQ TOPIC — GUARANTEED, every paper

A. The two exam circuits + operation

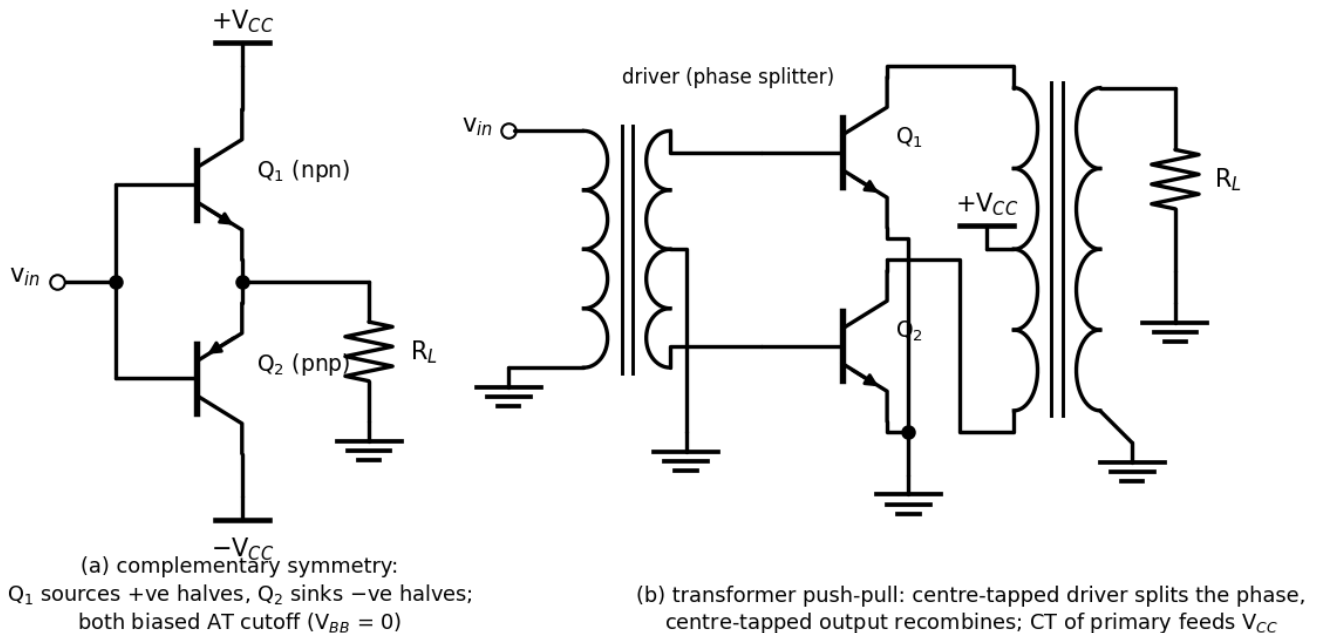


Fig. 6 — The two Class B realisations the PYQs ask for: (a) complementary-symmetry pair; (b) transformer-coupled push-pull (2082 Bhadra Q7 and 2081 Q10 name this one). In both, each device is biased AT cutoff and conducts exactly 180° . [rendered locally]

- **(a) Complementary symmetry:** matched npn + pnp, bases driven together, both $V_{BE} = 0$ at rest. The npn *sources* load current on positive half-cycles; the pnp *sinks* it on negative halves. No transformers, but needs complementary devices (and suffers crossover — §4.4).
- **(b) Transformer push-pull:** a centre-tapped driver transformer feeds the two npn bases in **antiphase** (phase splitting); each transistor amplifies its own half-cycle; the centre-tapped output transformer subtracts the two collector currents, recombining a full sine in the secondary. The primary CT feeds V_{CC} .

Class B: each device carries HALF-sinusoid pulses; the pair reconstructs the sine

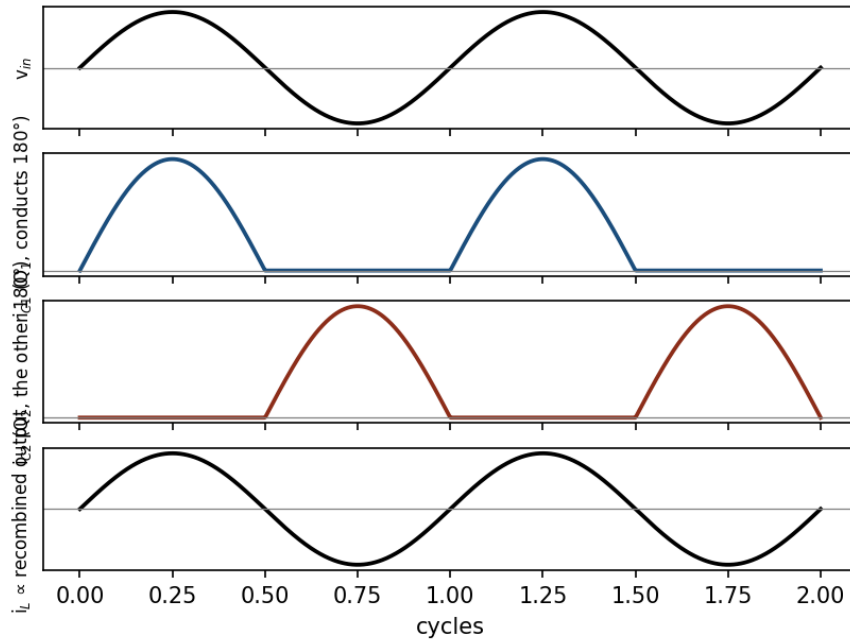


Fig. 7 — Computed conduction pattern: each collector carries half-sinusoid pulses; their difference through the output coupling reconstructs the sine. [computed]

B. The flagship efficiency derivation — PYQs 2081 Q10 [5], 2082 Bhadra Q7 [6]

- **Setup:** load voltage $v_L = V_p \sin \omega t$ across R_L ; peak device/load current $I_p = V_p/R_L$. The supply sees the two half-sine pulses back-to-back — i.e. a **full-wave-rectified** sine.
- **1. DC input:** average of a full-wave-rectified sine of peak I_p is $2I_p/\pi$, so $P_i(\text{dc}) = V_{CC} \cdot (2/\pi)(V_p/R_L) = 2V_{CC}V_p/(\pi R_L)$.
- **2. AC output:** $P_o(\text{ac}) = V_p^2/(2R_L)$.
- **3. Divide:**

$$\eta = P_o/P_i = [V_p^2/2R_L] / [2V_{CC}V_p/\pi R_L] = (\pi/4) \cdot (V_p/V_{CC}) \times 100\% \text{ — efficiency rises LINEARLY with drive}$$

- **4. Maximum — state the condition:** $V_p = V_{CC}$ (rail-to-rail swing, $V_{CE(\text{sat})} \approx 0$):

$$\eta_{\text{max}} = \pi/4 = 78.54\% \blacksquare$$

- **2082 Bhadra Q7's twist — “condition for minimum efficiency”:** since $\eta \propto V_p$, efficiency is **minimum ($\rightarrow 0$) as $V_p \rightarrow 0$** (idle and small signals) and maximum 78.5% at $V_p = V_{CC}$. Answer both directions in one sentence and you are safe against either intended reading.

C. Transistor dissipation — the calculus the maxima numerical needs

- **1.** What the devices burn is what the supply gives minus what the load gets: $P_{2Q} = P_i - P_o = 2V_{CC}V_p/(\pi R_L) - V_p^2/(2R_L)$.
- **2.** Maximise over drive: $dP_{2Q}/dV_p = 2V_{CC}/(\pi R_L) - V_p/R_L = 0 \Rightarrow V_p^* = 2V_{CC}/\pi \approx 0.636 \cdot V_{CC}$.
- **3.** Substitute back: $P_{2Q}(\text{max}) = 4V_{CC}^2/(\pi^2 R_L) - 2V_{CC}^2/(\pi^2 R_L)$:

$$P_{2Q}(\text{max}) = 2V_{CC}^2/(\pi^2 R_L) \text{ occurring at } V_p = 2V_{CC}/\pi \text{ — at PART drive, not full drive } (\approx 40\% \text{ of } P_o(\text{max}); \text{ half per transistor})$$

Class B powers vs drive — computed for the 2082 Baishakh Q10 values ($V_{CC} = 30\text{ V}$, $R_L = 16\ \Omega$)

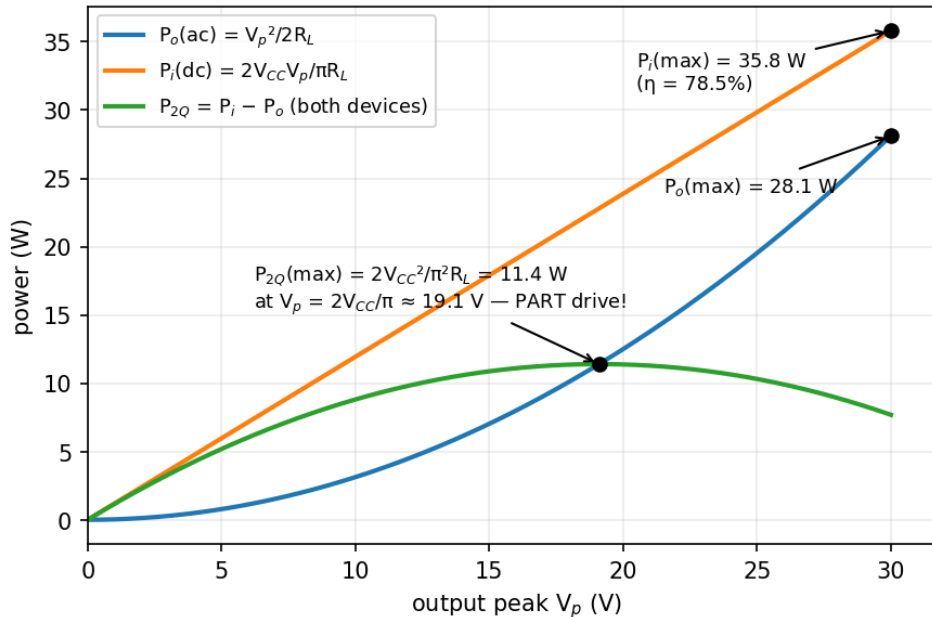


Fig. 8 — All three powers versus drive, computed for the 2082 Baishakh Q10 values ($V_{CC} = 30\text{ V}$, $R_L = 16\ \Omega$). The dissipation curve peaks at 19.1 V — the exact trap the question checks. [computed from the equations above]

★ **Worked PYQ 1 — 2083 Baishakh Q7 [6] (Boylestad's classic numbers)**

Given: Class B providing a 20 V peak signal to a 16 Ω load (speaker), supply $V_{CC} = 30\text{ V}$. Find the input power, output power and circuit efficiency.

- 1. $I_p = V_p/R_L = 20/16 = 1.25\text{ A} \Rightarrow I_{dc} = 2I_p/\pi = 2(1.25)/\pi = 0.796\text{ A}$
- 2. $P_i(dc) = V_{CC} \cdot I_{dc} = 30 \times 0.796 = \mathbf{23.9\text{ W}}$
- 3. $P_o(ac) = V_p^2/2R_L = 400/32 = \mathbf{12.5\text{ W}}$
- 4. $\eta = 12.5/23.9 \times 100 = \mathbf{52.4\%}$ ✓ sanity check with the formula: $(\pi/4)(20/30) = 52.36\%$ ✓

★ **Worked PYQ 2 — 2082 Baishakh Q10 [5] (the maxima version, same $V_{CC} = 30\text{ V}$, $R_L = 16\ \Omega$)**

Asked: maximum input power, maximum output power, and maximum power dissipation at the transistors.

- 1. $P_o(max) = V_{CC}^2/2R_L = 900/32 = \mathbf{28.1\text{ W}}$ (at $V_p = V_{CC} = 30\text{ V}$)
- 2. $P_i(max) = 2V_{CC}^2/(\pi R_L) = 1800/(16\pi) = \mathbf{35.8\text{ W}}$ (also at full drive; η there = $28.1/35.8 = 78.5\%$ ✓)
- 3. $P_{2Q}(max) = 2V_{CC}^2/(\pi^2 R_L) = 1800/(9.870 \times 16) = \mathbf{11.4\text{ W}}$, occurring at $V_p = 2V_{CC}/\pi \approx \mathbf{19.1\text{ V}}$ — **write that it peaks at part drive**; that remark is the discriminating mark (see Fig. 8).
- ⚠ Mistakes: $I_{dc} = I_p/\pi$ (that is a HALF-wave average — Class B's supply sees full-wave, $2I_p/\pi$) · computing $P_{2Q}(max)$ at $V_p = V_{CC}$ · mixing this class's formulas with Class A's · forgetting to say each device conducts 180°.

4.4-4.5 Class AB Operation and Its Biasing — PYQ 2082 Bhadra Q8 [6]

“Define crossover distortion in a class B amplifier and explain how class AB operation can be used to eliminate it.”

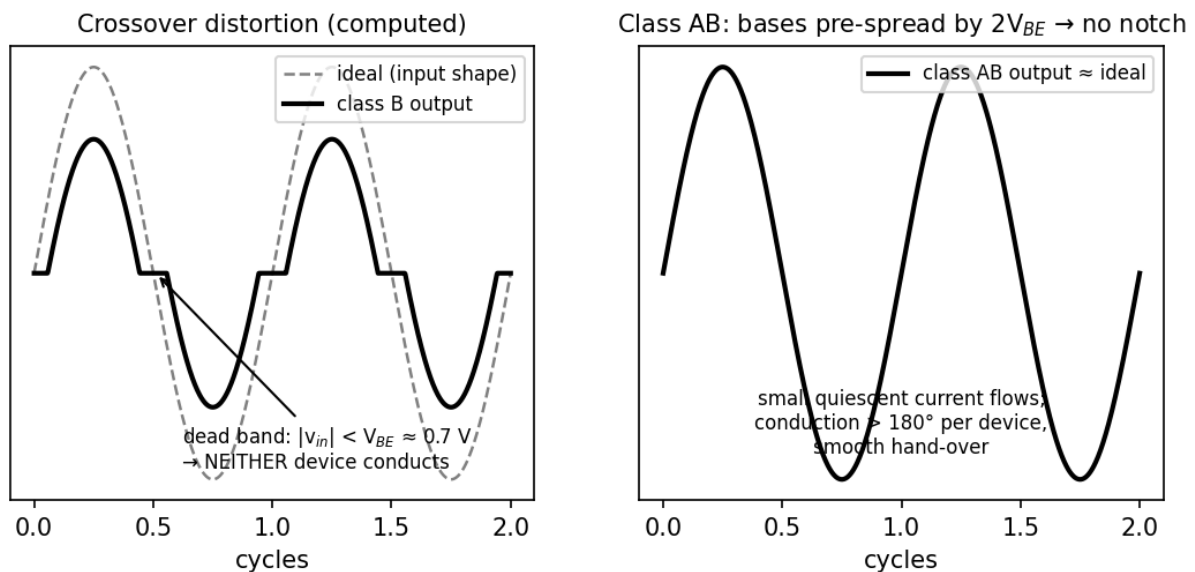


Fig. 9 — Left: computed Class B output — the transfer dead-band $|v_{in}| < V_{BE}$ notches every zero-crossing. Right: with class-AB pre-bias the notch is gone. [computed from the piecewise transfer law]

Crossover distortion — the definition [≈2 marks]

In true Class B both devices are biased exactly at cutoff, so for $|v_{in}| < V_{BE} \approx 0.7 \text{ V}$ **neither transistor conducts**: the output sits at zero for a moment around every zero-crossing. The resulting notch is odd-harmonic distortion, and — crucially — it is **worst for small signals**, where 0.7 V is a large fraction of the swing.

The Class AB cure [≈4 marks]

- **Idea**: hold the two bases apart by $\approx 2V_{BE}$ so that **both devices carry a small quiescent current** at rest. Each then conducts slightly more than 180° , and conduction hands over smoothly through the zero-crossing — no dead band.
- **Bias method 1 — two diodes** in the base chain: cheap, and the diode drops *track* V_{BE} *thermally* (both fall $\approx -2 \text{ mV}/^\circ\text{C}$), stabilising the quiescent current against thermal runaway — mention this tracking for a mark.
- **Bias method 2 — V_{BE} multiplier** (“rubber diode”): a transistor with a divider R_1 – R_2 across it forces $V_{CE} = V_{BE}(1 + R_1/R_2)$ — an *adjustable* spread, the standard trick inside IC output stages.
- **Bias method 3** — plain resistor divider: crude, poorly tracked; list it last.
- **Cost**: a small standby dissipation, so $\eta_{\max}$ sits slightly below 78.5% — the accepted price of linearity.

4.6 Power BJTs (device limits)

● **LOW** Large-area devices with low β (20–100). What matters is the ratings box: $I_{C(\max)}$, V_{CEO} , and $P_{D(\max)}$ quoted at 25°C case temperature with a **derating curve** above it. Junction temperature obeys the thermal ladder $T_J = T_A + \theta_{JA} \cdot P_D$, $\theta_{JA} = \theta_{JC} + \theta_{CS} + \theta_{SA}$ — a heat sink attacks the last term. Add **second breakdown** and the SOA (safe-operating-area) plot. 2–3 mark readiness.

4.7 Transformer-Coupled Push-Pull — the extra credit lines

- The centre taps do the phase **splitting** (driver) and **recombination** (output) — already drawn in Fig. 6(b).

- **Even harmonics cancel:** the two half-windings drive the core in opposite senses, so even-order terms (which are symmetric in the two halves) produce cancelling flux — push-pull output contains only odd residues. This line upgrades a [5] answer.
- **No DC core saturation:** the two quiescent currents magnetise the core in opposition. The same skeleton runs Class A or AB by moving the base bias.

4.8 Tuned (Class A) Amplifiers ★ REPEATED PYQ TOPIC — PYQs 2081 Q11 [4] (circuit + response + 3 dB BW) and 2082 Baishakh Q7 [4] (derive BW)

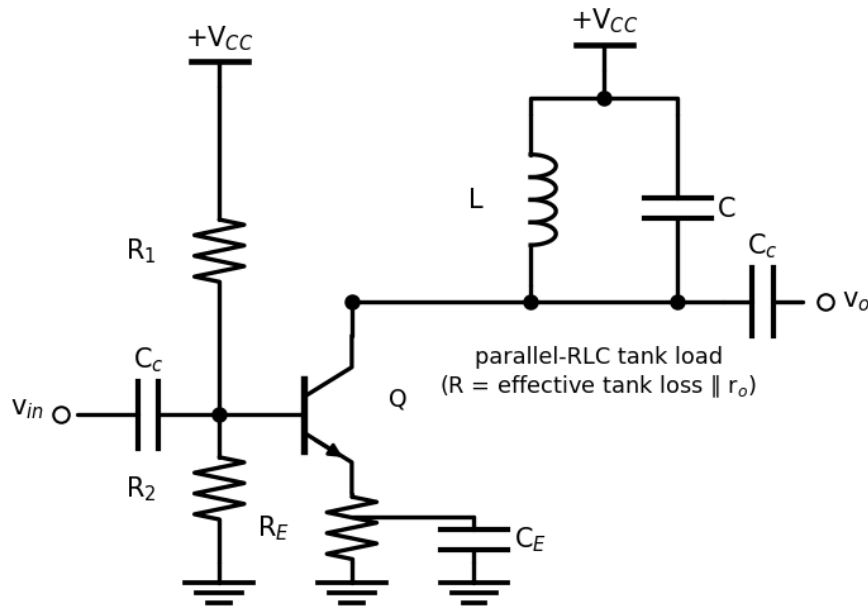


Fig. 10 — Class A tuned amplifier: an ordinary CE stage whose collector load is a parallel-RLC tank ($R = \text{tank loss} \parallel r_o$). Bias is plain Class A — the tank only shapes the FREQUENCY response. [rendered locally]

Derivation — do it as this five-line chain

- **1.** Gain: $A(j\omega) = -g_m \cdot Z_{\text{tank}}(j\omega)$ with $Y_{\text{tank}} = 1/R + j(\omega C - 1/\omega L)$.
- **2.** Factor R out: $Z = R / [1 + jR(\omega C - 1/\omega L)] = R / [1 + jQ(\omega/\omega_0 - \omega_0/\omega)]$, defining

$$\omega_0 = 1/\sqrt{LC} \quad (f_0 = 1/2\pi\sqrt{LC}) \quad Q = \omega_0 CR = R/(\omega_0 L)$$

- **3.** At ω_0 the tank is purely resistive (the L and C currents cancel) $\Rightarrow A_{\text{max}} = -g_m R$ — the peak of the response.
- **4.** Half-power (−3 dB) points: $|1 + jQ(\cdot)| = \sqrt{2} \Rightarrow Q(\omega/\omega_0 - \omega_0/\omega) = \pm 1$. Multiply by $\omega\omega_0/Q$: $\omega^2 \mp (\omega_0/Q)\omega - \omega_0^2 = 0$ — two quadratics whose positive roots are ω_1 (lower) and ω_2 (upper).
- **5.** Subtracting the two positive roots, the $\pm(\omega_0/Q)$ terms add: $\omega_2 - \omega_1 = \omega_0/Q$. Hence:

$$\text{BW} = f_2 - f_1 = f_0/Q = 1/(2\pi RC) \quad (\text{using } Q = \omega_0 CR) \quad \cdot \text{ also } f_0 = \sqrt{f_1 f_2} \text{ — the geometric centre}$$

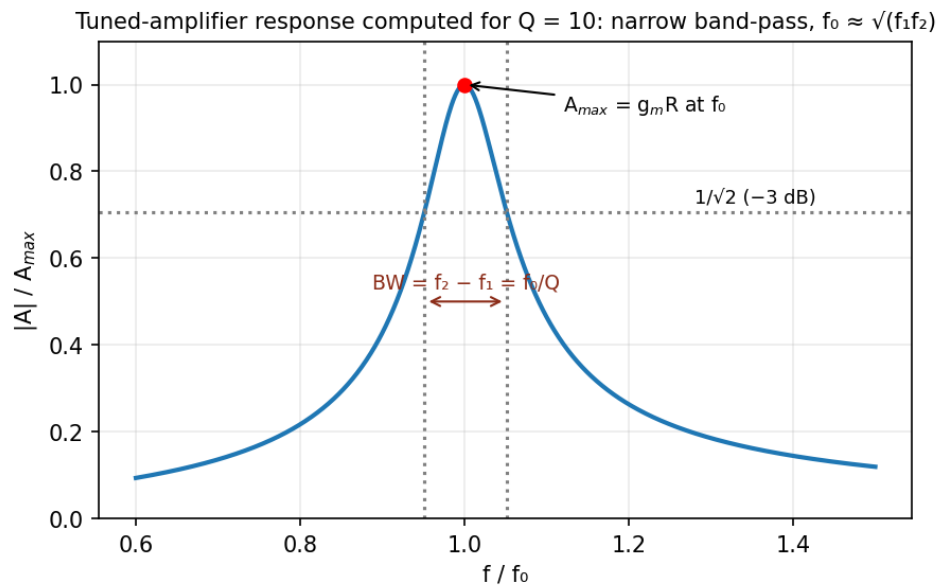


Fig. 11 — The band-pass response computed from the $Z(j\omega)$ expression for $Q = 10$, with the -3 dB points and $BW = f_0/Q$ marked — exactly the sketch the 2081 Q11 answer needs. [computed]

- **Selectivity** $S = f_0/BW = Q$ — one line. Bridge to Class C: a tuned Class B/C stage still outputs a clean sine because the tank's **flywheel effect** rings through the gaps between current pulses — that is how $>90\%$ RF stages exist.
- ⚠ Mistakes: $BW = f_0 \cdot Q$ (upside-down) · forgetting $A_{max} = g_m R$ (not $g_m \omega_0 L$) · calling the stage 'Class C' — the PYQs say *Class A tuned*: normal linear bias, tuned load.

Chapter 4 Final Revision

Conceptual summary (read this the night before)

Output stages trade linearity for efficiency through the conduction angle. Series-fed Class A idles at I_{CQ} , burns $V_{CC}I_{CQ}$ even in silence, and caps at 25%. Transformer coupling removes the DC resistor loss and swings V_{CE} to $2V_{CC}$: $\eta = 50[(V_{CEmax} - V_{CEmin})/(V_{CEmax} + V_{CEmin})]^2$ %, maximum 50%. Class B biases both devices at cutoff — each conducts 180° , the supply sees full-wave-rectified current $2I_p/\pi$ — giving $\eta = (\pi/4)(V_p/V_{CC})$, maximum 78.5% at rail swing, with device dissipation peaking at PART drive $V_p = 2V_{CC}/\pi$ ($P_{2Q,max} = 2V_{CC}^2/\pi^2 R_L$). Its price is the crossover notch (dead band $|v_{in}| < V_{BE}$), cured by Class-AB pre-bias of $2V_{BE}$ via tracking diodes or a V_{BE} multiplier. Push-pull transformer stages phase-split/recombine and cancel even harmonics. Power BJTs live inside the thermal ladder $T_J = T_A + \theta_{JA} P_D$ and the SOA. A parallel-RLC collector load makes the tuned amplifier: $f_0 = 1/2\pi\sqrt{LC}$, $A_{max} = -g_m R$, and **BW** = $f_0/Q = 1/2\pi RC$.

Important diagrams checklist

Figure	Priority	Marked in PYQ as
Fig. 6 — Class B: complementary + transformer push-pull	● HIGH	2082 Bh Q7, 2081 Q10 ★
Fig. 8 — Class B power curves (P_{2Q} peak at part drive)	● HIGH	2082 Ba Q10
Fig. 4 + 5 — transformer-A circuit + DC/AC load lines	● HIGH	2083 Q8 ★, 2081 Q9
Fig. 9 — crossover notch + AB fix	● HIGH	2082 Bh Q8
Fig. 10 + 11 — tuned amp + band-pass response	● HIGH	2081 Q11 ★, 2082 Ba Q7
Fig. 2 + 3 — series-fed circuit + centred-Q load line	● MEDIUM	2082 Ba Q9
Fig. 7 — half-sine conduction waveforms	● MEDIUM	supports every Class B answer
Fig. 1 — conduction-angle class comparison	● MEDIUM	classification lead-ins

Formula sheet

Series A: $P_i = V_{CC}I_{CQ}$ · $P_o = V_p^2/2R_L$ · $\eta_{max} = 25\%$

Transformer A: $\eta = 50[(V_{CEmax} - V_{CEmin})/(V_{CEmax} + V_{CEmin})]^2$ % = $50(V_p/V_{CC})^2$ % · $\eta_{max} = 50\%$ · swing $\rightarrow 2V_{CC}$

Class B: $I_{dc} = 2V_p/\pi R_L$ · $P_i = 2V_{CC}V_p/\pi R_L$ · $P_o = V_p^2/2R_L$ · $\eta = (\pi/4)(V_p/V_{CC})$ · $\eta_{max} = 78.5\%$

Class B maxima: $P_o(max) = V_{CC}^2/2R_L$ · $P_i(max) = 2V_{CC}^2/\pi R_L$ · $P_{2Q}(max) = 2V_{CC}^2/\pi^2 R_L$ at $V_p = 2V_{CC}/\pi$

AB: base spread $2V_{BE}$ · V_{BE} -multiplier: $V = V_{BE}(1 + R_1/R_2)$ · **Thermal:** $T_J = T_A + \theta_{JA} P_D$

Tuned: $f_0 = 1/2\pi\sqrt{LC}$ · $Q = \omega_0 CR = R/\omega_0 L$ · $A_{max} = -g_m R$ · **BW** = $f_0/Q = 1/(2\pi RC)$ · $f_0 = \sqrt{f_1 f_2}$

Priorities / mistakes / derivation list

- **HIGH** Class B: η derivation AND both numerical templates (★4/4) · transformer-A general/max η + numerical (★2/4) · tuned BW derivation (★2/4) · crossover/AB story (1/4 but a full 6-marker when it lands).
- **MEDIUM** Series-fed 25% derivation · P_{2Q} calculus · push-pull even-harmonic cancellation.
- **LOW** Classification table details · power-BJT thermal/SOA · Class C beyond the flywheel sentence.
- ⚠ Top mistakes: $I_{dc} = I_p/\pi$ instead of $2I_p/\pi$ · swing capped at V_{CC} in transformer coupling (it reaches $2V_{CC}$) · P_{2Q} evaluated at full drive · class formulas cross-contaminated · **BW** = $f_0 \cdot Q$ · omitting '180° each' from Class B answers.

- Derivations to rehearse: series-A 25% (4 lines) · transformer-A general η · Class-B η (the flagship — ≤ 8 lines) · $P_{20}(\text{max})$ via $dP/dV_p = 0$ · tuned BW via the two quadratics. Results to hard-memorise: 25 / 50 / 78.5% · $2V_{CC}/\pi$ · even harmonics cancel.

Complete Chapter 4 PYQ Bank (wording preserved)

Year / Exam	Q	Question (condensed, wording preserved)	Mark s	Type
2083 Baishakh (Back)	7	For a class B amplifier providing a 20 V peak signal to a 16 Ω load (speaker) and a power supply of $V_{CC} = 30$ V, determine the input power, output power, and circuit efficiency.	6	Numerical
2083 Baishakh (Back)	8	Draw the circuit diagram and the characteristic curve of a transformer-coupled class A amplifier and derive its general and maximum efficiency.	7	Derivation
2082 Bhadra (Regular)	7	Draw the circuit diagram of transformer coupled Class B amplifier. Determine the condition for minimum efficiency of Class B amplifier.	6	Theory + Derivation
2082 Bhadra (Regular)	8	Define crossover distortion in a class B amplifier and explain how class AB operation can be used to eliminate it.	6	Theory
2082 Baishakh (Back)	7	Derive the expression of bandwidth of class A tuned amplifier.	4	Derivation
2082 Baishakh (Back)	9	Derive general efficiency of series fed class A amplifier.	4	Derivation
2082 Baishakh (Back)	10	For a class B amplifier using a supply of $V_{CC} = 30$ V and driving a load of 16 Ω , determine the maximum input power, maximum output power and maximum power dissipation at transistors.	5	Numerical
2081 Ashwin (Regular)	9	Calculate the efficiency of a transformer coupled Class A amplifier for a supply of 12 V DC and output of $V_p = 12$ V and 6 V.	4	Numerical
2081 Ashwin (Regular)	10	Draw the circuit diagram and the characteristic curve of a transformer coupled class B push-pull amplifier and derive its general efficiency.	5	Derivation
2081 Ashwin (Regular)	11	Draw the circuit diagram of class A tuned amplifier with its frequency response. Derive its 3 dB bandwidth.	4	Derivation

PYQ → What to Study Mapping

PYQ	Topic	Notes section(s)	Required skill
2083 Q7 [6]	Class B numerical	§4.3-D (template §4.3-B)	$I_{dc} = 2I_p/\pi \rightarrow P_i, P_o, \eta$
2083 Q8 [7]	Transformer A	§4.2-B (Figs. 4-5)	circuit + BOTH load lines + general $\eta \rightarrow 50\%$
2082 Bh Q7 [6]	Class B transformer + η condition	§4.3-A(b), §4.3-B, §4.7	circuit + $\eta \propto V_p$ argument, both extremes
2082 Bh Q8 [6]	Crossover / AB	§4.4-4.5 (Fig. 9)	definition + dead-band cause + $2V_{BE}$ cure + bias methods
2082 Ba Q7 [4]	Tuned BW	§4.8 (Figs. 10-11)	$Z_{tank} \rightarrow Q \rightarrow$ two quadratics $\rightarrow BW = f_0/Q = 1/2\pi RC$
2082 Ba Q9 [4]	Series-fed A	§4.2-A (Figs. 2-3)	P_i constant, $P_o = V_p^2/2R_L \rightarrow 25\%$
2082 Ba Q10 [5]	Class B maxima	§4.3-C, E (Fig. 8)	three maxima + part-drive dissipation remark
2081 Q9 [4]	Transformer A numerical	§4.2-B worked example	$\eta = 50(V_p/V_{CC})^2$ % at both amplitudes
2081 Q10 [5]	Push-pull η	§4.3-A(b), §4.3-B, §4.7	circuit + waveforms + $(\pi/4)(V_p/V_{CC})$
2081 Q11 [4]	Tuned amp	§4.8	circuit + response sketch + 3 dB derivation

Final Coverage Audit

- ✓ Syllabus 4.1–4.8 all present in order; classification, power BJTs and Class-C/flywheel included at readiness depth though never asked directly (coverage rule).
- ✓ References honoured: Boylestad (every efficiency formula and both numericals follow his worked examples — same 30 V/16 Ω numbers), Sedra-Smith (classification, AB biasing, V_{BE} multiplier),

Millman/Floyd corroboration. Conventions flagged: $P_i(\text{dc})$ means total supply power; V_p is the load peak.

- ✓ All 10 PYQs embedded in their sections AND repeated in the bank + mapping; both Class-B numericals solved with every arithmetic step; the ambiguous 'minimum efficiency' wording answered both ways; the P_{2Q} part-drive trap called out explicitly and shown graphically in Fig. 8.
- ✓ ★ Repeated topics flagged (Class B 4/4, transformer-A 2/4, tuned 2/4); ● priorities from PYQ evidence; 11 figures rendered locally — load lines, waveforms and the power curves computed from the governing equations using the exam papers' own values ($V_{CC} = 30\text{ V}/16\ \Omega$, $V_{CC} = 12\text{ V}$), per the fetch report (0/11 from web).

End of Chapter 4 notes. Companions: Chapter 1 (BJT, markdown), Chapters 2–3 (PDF, same series); Chapter 5 pending the same upgrade.